

Capstone Project – NLP Course (open book assignment)

Title

End-to-End NLP System Development (Choice-Based Capstone)

Problem Statement

Design, implement, and deploy a **real-world NLP system** integrating multiple techniques learned in this course (preprocessing, classical ML, embeddings, RNNs, Transformers). You will choose one of three project tracks and deliver a production-ready prototype with evaluation, visualization, and deployment.

Project Tracks

1. **Sentiment Analysis Dashboard (Business Intelligence Focus)**

- Real-time sentiment analysis of reviews/tweets.
- Compare classical ML, embeddings+RNN, and transformer models.
- Build a **Streamlit dashboard** with visualizations, attention heatmaps, and comparison plots.

2. **CV/Resume Parser (HR Tech Focus)**

- Extract structured information (name, skills, experience, education) from resumes.
- Classify resumes into job categories (ML classification).
- Build a **searchable interface**: query candidates by skills → ranked list.

3. **Content Moderation System (Trust & Safety Focus)**

- Detect hate speech, toxic comments, and offensive language.
- Multi-label classification (toxic, offensive, hate speech, harassment, etc.).
- Build a **moderation dashboard** with explainability: highlight toxic words (using attention visualization).

Detailed Tasks

Phase 1: Data Collection & Preprocessing (10%)

- Collect or download dataset (IMDB, Kaggle Resumes, Hate Speech datasets).
- Apply text cleaning pipelines (NLTK + spaCy).
- Build reusable preprocessing module.

Phase 2: Baseline Models (15%)

- Classical ML models (Naive Bayes, Logistic Regression with TF-IDF).
- Evaluate baselines for accuracy, precision, recall.

Phase 3: Deep Learning Models (30%)

- Word2Vec/GloVe embeddings + LSTM/GRU for classification.
- Fine-tune transformer models (BERT, DistilBERT, RoBERTa).
- Compare across multiple architectures.

Phase 4: Explainability & Visualization (20%)

- Use **attention visualization** for Transformers.
- Show feature importance for ML baselines.
- t-SNE embedding visualization for clustering.

Phase 5: Deployment & UI (15%)

- Build **Streamlit app** with:
 - Text input → classification output.
 - Confidence scores.
 - Attention/embedding visualizations.
- Export model to TorchScript/ONNX for efficiency.

Phase 6: Report & Presentation (10%)

- Document dataset, experiments, results, visualizations.
- Compare classical ML vs Deep Learning vs Transformer performance.
- Reflect on scalability, limitations, and improvements.

Deliverables

- Code repository with modular structure.
- Trained models + exported ONNX/TorchScript.
- Streamlit app with working demo.
- Technical report (8–10 pages).
- 5-min recorded video demo (optional, extra credit).

Evaluation Rubric (100 points)

- Data preprocessing & baseline models: 15
- Deep learning + transformers: 30
- Explainability & visualization: 20
- Deployment: 15
- Report quality: 10
- Innovation / extra features: 10

Target Outcomes

- **Sentiment Dashboard:** >90% accuracy with BERT, interactive dashboard.
- **CV Parser:** $\geq 85\%$ F1-score for skill extraction, searchable interface.
- **Content Moderation:** >85% macro-F1, reliable detection of multiple labels.
